

Solution

MOTION-01

Class 09 - Science

1.

(b) increasing with time

Explanation: Acceleration is directly proportioned to time, it varies linearly with time and increases with respect to time.

2.

(d) 4 sec

Explanation: Using kinematic equation, $v = u + a \times t$, when ball reaches to top of its trajectory, its velocity becomes zero. Ball is thrown against gravity of earth, therefore $a = -g$, $u = 20$ m/s, we can find time of ascend as

$$0 = 20 - 10 \times t$$

$$20 = 10 \times t$$

$$t = \frac{20}{10} = 2 \text{ sec}$$

Time of ascend is same as time of descend = 2 sec.

Time of flight = time of ascend + time of descend

Therefore time of flight = 4 sec.

3.

(b) Displacement

Explanation: The area under the velocity-time graph gives the distance (magnitude of displacement) which has the unit is metre (m).

4.

(b) $\frac{L}{v} + \frac{v}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$

Explanation: Velocity increases from 0 to v :

We know that $v = u + at$

$$\text{Here, } u = 0, v = v, a = \alpha \therefore t = \frac{v}{\alpha}$$

$$\text{and } s = ut + \frac{1}{2} at^2; s_1 = \frac{v^2}{2\alpha}$$

Velocity decreases from v to 0:

$$\text{Here, } u = v, v = 0, a = -\beta$$

Using $v = u + at$

$$\Rightarrow t = \frac{v}{\beta}$$

$$\text{and } s = ut + \frac{1}{2} at^2; s_2 = \frac{v^2}{2\beta}$$

So, distance travelled during acceleration and retardation,

$$d = s_1 + s_2 = \frac{v^2}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$$

Thus, distance travelled during constant velocity

$$S_3 = L - \frac{v^2}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$$

So, time taken to travel this distance

$$t = \frac{L - \frac{v^2}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)}{v} = \frac{L}{v} - \frac{v}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$$

Hence, total time taken to cover distance L

$$= \left(\frac{v}{\alpha} \right) + \left[\frac{L}{v} - \frac{v}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) \right] + \left(\frac{v}{\beta} \right)$$

$$= \frac{L}{v} + \frac{2}{v} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$$

5. (a) uniform acceleration

Explanation: Relation among velocity, distance, time, and acceleration is called equations of motion. There are three equations of motion. Equation of motion can be used for a body having uniform acceleration.

6.

(b) $3\pi : 2\sqrt{2}$

Explanation: $r' = r\sqrt{2}$ displacement

$$\text{distance} = \frac{3\pi r}{2}$$

$$\text{ratio} = \frac{\frac{3\pi r}{2}}{r\sqrt{2}}$$

$$\text{ratio} = 3\pi : 2\sqrt{2}$$

7.

(d) 6.36 km h^{-1} , 0

Explanation: Time taken by the man to go from his home to the market, $t_1 = \frac{5.5 \text{ km}}{5.5 \text{ km h}^{-1}} = 1.0 \text{ h}$

Time taken by the man to return back from the market to his home, $t_2 = \frac{5.5 \text{ km}}{7.5 \text{ km h}^{-1}} = 0.73 \text{ h}$

$\therefore$ Total time taken = $t_1 + t_2 = 1.0 \text{ h} + 0.73 \text{ h} = 1.73 \text{ h} = 140 \text{ min}$

In $t = 0$ to 104 min , total distance travelled = $5.5 \text{ km} + 5.5 \text{ km} = 11 \text{ km}$

Displacement = 0 (As the boy returns back home)

$\therefore$ Average speed

$$= \frac{\text{Distance travelled}}{\text{Time taken}} = \frac{11 \text{ km}}{1.73 \text{ h}} = 6.36 \text{ km h}^{-1}$$

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Time taken}} = 0$$

8.

(c) 13.5 km h^{-1}

Explanation: Let, the boy runs with speed v for next 20 min

Given that, average speed = 12 km/h^{-1}

As, average speed = $\frac{\text{Total distance covered}}{\text{Total time taken}}$

$$12 = \frac{\text{Total distance covered}}{\frac{(10+20)}{60} \text{ h}}$$

$$\Rightarrow \text{Total distance covered} = 12 \times \frac{30}{60} = 12 \times \frac{1}{2} = 6 \text{ km}$$

$$\text{Distance covered in } 10 \text{ min} = 9 \times \frac{10}{60} = 1.5 \text{ km}$$

$$\text{Remaining distance} = 6 - 1.5 = 4.5 \text{ km}$$

For next 20 min , he should run with speed

$$v = \frac{4.5}{\frac{20}{60}} = 4.5 \times 3 = 13.5 \text{ km h}^{-1}$$

9. (a) Negative

Explanation: If a moving body comes to rest, then its acceleration is negative, as it's velocity becomes zero. therefore the acceleration is also zero.

10.

(b) zero

Explanation: The average velocity for the entire swing would be zero because its final position and initial position are identical.

11.

(b) -0.29 m s^{-1}

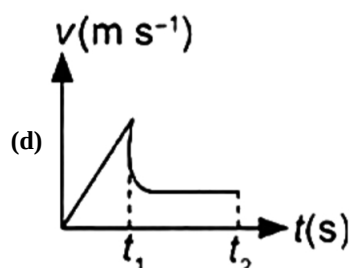
Explanation: During time interval $t = 2 \text{ s}$ to $t = 9 \text{ s}$, six dance steps namely AB, BC, CD, DE, EF and FG have been demonstrated by the dancer.

$$\text{Total displacement} = (0 - 2) = -2 \text{ m}$$

$$\text{Total time taken} = 9 - 2 = 7 \text{ s}$$

$$\text{Average velocity} = \frac{\text{total displacement}}{\text{total time taken}} = \frac{-2}{7} = -0.29 \text{ m s}^{-1}$$

12.



Explanation: When the parachutist jumps off the plane his vertical speed is 0 m/s . The speed will increase constantly due to

the gravitational field strength of the Earth, which makes the parachutist accelerate uniformly downwards. At this point, choice B and choice D are eliminated because the two graphs show a deceleration.

When the parachute opens, there is a sudden strong opposing air resistance acting against the parachutist, which causes the speed of the parachutist to drop drastically and eventually reach a constant speed known as terminal velocity.

13.

(b) equal or less than 1

Explanation: We know, the magnitude of displacement is the straight line distance between the initial point and the final point of the path of the moving body.

Displacement could be negative, positive, or zero. but the distance is always positive or zero when there is no motion.

So, the ratio of displacement to distance for a moving object is equal to less than 1.

14.

(d) Velocity

Explanation: The slope of a position graph represents the velocity of the object. The steeper the slope is, the faster the motion is changing. So the value of the slope at a particular time represents the velocity of the object at that

instant. Average velocity can be calculated from a position-time graph as the change in position divided by the corresponding change in time.

15. (a) It has only magnitude and no specific direction

Explanation: Distance is the length of path covered by a moving object in the given time irrespective of direction. Distance has only magnitude and no direction.

16.

(b) the acceleration

Explanation: The area under a velocity-time graph represents the distance covered and the gradient of a velocity-time graph represents the acceleration.

17.

(b) 5:9

Explanation: Average speed = $\frac{500}{10} = 50$ km/hr

Ratio of average speed to maximum speed = 50 : 90 = 5:9

18.

(c) 0.5 km/hr

Explanation: Distance = speed $\times$ time

Distance travelled in first 2 min = $7.5 \times \frac{2}{60} = 0.25$ km

Distance travelled in last 2 min = $7.5 \times \frac{2}{60} = 0.25$ km

Total distance = 0.25 + 0.25 = 0.5 km

Total time = 2 + 2 + 56 = 60 min = 1 hr

Average speed = $\frac{0.5}{1}$

= 0.5 km/hr

19.

(b) 1 : 1

Explanation: Both the cars make a complete circle in the same time T, which means that angular speeds of both the cars are same.

$$\therefore \omega = \frac{v}{r} = \frac{2\pi r}{Tr} = \frac{2\pi}{T}$$

20.

(d) 22 s

Explanation: Time taken when he falls 80 m in the air is t_1 , and $U_1 = 0$, $a_1 = 10 \text{ m s}^{-2}$

$$\therefore d = U_1 t_1 + \frac{1}{2} a_1 t_1^2 \text{ or } 80 = 0 + \frac{1}{2} \times 10 \frac{1}{2} t_1^2$$

$$u_1 = 0$$

$$a_1 = 10 \text{ m s}^{-2}$$

$$t_1 = ?$$

$$u_2 = v_1$$

$$a_2 = 2 \text{ m s}^{-2}$$

$$t_2 = ?$$

$$u_2 = 4 \text{ m s}^{-1}$$

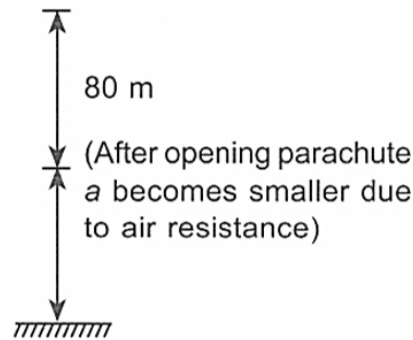
$$\text{or } t_1^2 = 16; t_1 = 4 \text{ s} \Rightarrow v_1 = u_1 + a_1 t_1$$

$$v_1 = 0 + 10 \times 4 = 40 \text{ m s}^{-1}$$

$$\text{Since } v_1 = u_2 \therefore u_2 = 40 \text{ m s}^{-1}$$

Let the time taken after he opens up his parachute and before reaches the ground is t_2 .

$$v_2 = 4 \text{ m s}^{-1}, a_2 = -2 \text{ m s}^{-2}$$



$$v_2 = u_2 + a_2 t_2 \Rightarrow 4 = 40 + (-2) \times t_2$$

$$2t_2 = 36; t_2 = 18 \text{ s}$$

$$\text{Total time} = 4 \text{ s} + 18 \text{ s} = 22 \text{ s}$$

21.

$$(d) v_1 > v_2$$

Explanation: In time interval 1, magnitude of net displacement is more as compared to that time interval 2 i.e., $\Delta x_1 > \Delta x_2$

$$\therefore v_1 > v_2$$

22.

$$(c) \frac{u + v}{2}$$

Explanation: Average velocity is the ratio of total displacement or total distance travelled by a body in a given interval of time.

23.

(d) its velocity changes but speed remain the same

Explanation: Speed remains the same as it is 'uniform' circular motion and velocity changes at every point because of direction changes. The tendency of the object at every point is to go tangential to its position.

24. (a) 50

Explanation: Distance travelled in t^{th} second of uniformly accelerated motion is

$$S_t = u + \frac{a}{2}(2t - 1) \dots (i)$$

Distance travelled in $(t + 1)^{\text{th}}$ second can be written as

$$S_{t+1} = u + \frac{a}{2}[2(t + 1) - 1]$$

$$\text{or } S_{t+1} = u + \frac{a}{2}(2t + 1) \dots (ii)$$

$$s_t + s_{t+1} = 100 \text{ cm (given)}$$

$$u + \frac{a}{2}(2t - 1) + u + \frac{a}{2}(2t + 1) = 100 \text{ [Using (i) and (ii)]}$$

$$\text{or } 2u + 2at = 100 \text{ or } u + at = 50;$$

$$\therefore v = 50 \text{ cm s}^{-1}$$

25.

$$(d) 2.75 \text{ m s}^{-1}$$

Explanation: Distance covered for dance step CD = 4 m

Time taken for CD is greater than 1 s but less than 2 s

Since, average speed = $\frac{\text{total distance}}{\text{total time}}$

Therefore, average speed is less than 4 m s^{-1} but greater than 2 m s^{-1} . Hence, option (C) is the only possibility.

26.

(c) m

Explanation: The area under the velocity-time graph gives the distance (magnitude of displacement) which has the unit: metre (m)

27.

(d) Both (i) and (ii) are correct

Explanation:

i. As the part BC of the speed-time graph has maximum slope, so acceleration is maximum during this interval.

Maximum acceleration = Slope of BC

$$= \frac{(50-20)\text{kmh}^{-1}}{(1.00-0.75)\text{h}} = \frac{30}{0.25} = 120 \text{ km h}^{-2}$$

ii. Distance covered from 0.75 h to 1.00 h

= Area of trapezium BEFC

$$= \frac{1}{2} [\text{BE} + \text{CF}] \times \text{EF}$$

$$= \frac{1}{2} [20 + 50] \times (1.00 - 0.75) \text{ km} = 8.75 \text{ km}$$

28.

(b) All of these

Explanation: Displacement is the shortest distance travelled so it can be positive or negative or zero.

29.

(b) zero

Explanation: The slope is zero (because the slope for the velocity time graph gives the acceleration).

30. (a) acceleration

Explanation: We can find out the value of acceleration from the slope of the velocity-time graph of a moving body.

Acceleration = $\frac{\text{Change in velocity}}{\text{time}}$ = Slope of the velocity-time graph provided.

31. (a) 25.5 ms^{-1}

Explanation: Let the acceleration of the car is = a

And the distance between A and B is = d

$$v^2 - u^2 = 2ad$$

$$v = 30\text{m/s and } u = 20\text{m/s}$$

$$2ad = 30^2 - 20^2$$

$$Ad = \frac{(900-400)}{2} = 250$$

When the car is at the mid point of AB then let the speed of the car is v_1

$$v_1^2 - 20^2 = 2a \left(\frac{d}{2}\right)$$

$$v_1^2 = ad + 400 = 250 + 400 = 650$$

therefore v_1 is = 25.4950m/s .

32.

(c) 2.4 ms^{-1}

Explanation: The average distance covered in unit time by a moving object is called average speed. The average speed is the ratio of total distance covered and total time taken.

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}} = \frac{10+20+30}{5+10+10} = \frac{60}{25} = 2.4 \text{ ms}^{-1}$$

33.

(c) falls back in his hand

Explanation: A person sitting in the truck projected a ball vertically upwards. The ball will fall back in his hand as the ball and the truck are moving at the same speed.

34. (a) Displacement

Explanation: The area enclosed by the velocity-time graph and time axis gives the magnitude of displacement.

$$\therefore \text{velocity} = \frac{\text{displacement}}{\text{Time}}$$

$$\Rightarrow \text{Velocity} \times \text{time} = \text{displacement}$$

35.

(c) in accelerated motion

Explanation: In a merry go round, the speed is constant but the velocity is not constant because its direction will be changing i.e., there is acceleration in the motion. So the boy is in accelerated motion.

36.

(b) 9×10^{10} m

Explanation: Time = 5 minutes = $5 \times 60 = 300$ sec.

Distance = speed $\times$ time

$$= 3 \times 10^8 \text{ m/s} \times 300 = 9 \times 10^{10} \text{ m}$$

37.

(b) Positively accelerated

Explanation: If a body starts from rest, it starts moving. that means the change in velocity is positive. That means there is a **POSITIVE** acceleration.

38.

(b) Velocity changes non-linearly if acceleration is changing linearly.

Explanation: For the given graph, $a = kt$, where k is some positive constant. So, velocity $v = u + at = u + (kt)t = 4 + kt^2$ i.e., varies non-linearly.

39. (a) constant

Explanation: Displacement is proportional to square of time

s is displacement and t is time then,

$$s \propto t^2$$

$$s = kt^2$$

$$\text{velocity}(v) = 2kt$$

$$\text{Acceleration}(a) = 2k = \text{constant.}$$

So, the motion of the body is constant acceleration.

40.

(d) Statement 1 is false but statement 2 is true.

Explanation: Let the distance travelled by the vehicle before it stops be d_s , then substituting $v = 0$, $x = d_s$ (stopping distance)

$$\text{in } v^2 = u^2 - 2ax, \text{ we get } d_s = \frac{u^2}{2a}.$$

Thus, the stopping distance is proportional to the square of the initial velocity.

So, if the initial velocity is reduced to half of its value, the stopping distance decreases by a factor of four.

41.

(c) Displacement has magnitude as well as specific direction

Explanation: The shortest distance between the initial point and the final point is called displacement. Displacement has both magnitude and direction while distance has only magnitude.

42.

(b) 338 m s^{-2}

Explanation: Here height from which ball is dropped,

$$h = 20 \text{ m}$$

$$\text{Using } v^2 = u^2 + 2gh; v^2 = 2gh \text{ (}\therefore u = 0\text{)}$$

$$\text{or } v = \sqrt{2gh} \text{ (downward)}$$

Ball rebounds to a height, $h' = 10 \text{ m}$

$$\text{So, } 0^2 = v'^2 - 2gh' \text{ or } v' = \sqrt{2gh'} \text{ (upward)}$$

Acceleration of the ball,

$$a = \frac{(v' - v)}{t} = \frac{[\sqrt{2gh'} - (-\sqrt{2gh})]}{t}$$

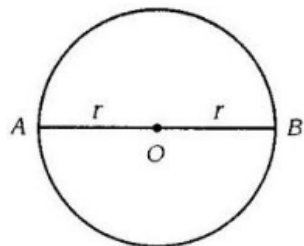
$$\text{or } a = \frac{[\sqrt{2 \times 9.8 \times 10} + \sqrt{2 \times 9.8 \times 20}]}{0.1} \simeq 338 \text{ ms}^{-2}$$

43. (a) $2r$

Explanation:

Given, after half the circle, the particle will reach the diametrically opposite point i.e., from point A to point B. And we know displacement is shortest path between initial and final point.

Displacement after half circle = AB = OA + OB [$\because$ Given, OA and OB = r]



$$= r + r = 2r$$

Hence, the displacement after half circle is $2r$.

44.

(d) $7.2 \text{ m}^{-1} \text{ s}^2$

Explanation: For motion with uniform acceleration a_1 :

From $v = u + at$

$$v = a_1 t_1 (\because u = 0), \therefore t_1 = \frac{v}{a_1} \dots (i)$$

and from $s = ut + \frac{1}{2}at^2$,

Using eqn. (i)

$$s_1 = \frac{1}{2} a_1 \left(\frac{v}{a_1} \right)^2 = \frac{v^2}{2a_1} \dots (ii)$$

For motion with uniform retardation a_2 :

From $v = u + at$

$$v = a_2 t_2 (\because v = 0, u = v, a = -a_2)$$

$$t_2 = \left(\frac{v}{a_2} \right) \dots (iii)$$

and from $s = ut + \frac{1}{2}at^2$

$$s_2 = v \frac{v}{a_2} + \frac{1}{2} (-a_2) \cdot \frac{v^2}{a_2^2}$$

$$s_2 = \frac{v^2}{a_2} - \frac{v^2}{2a_2} = \frac{v^2}{2a_2} \dots (iv)$$

Given, $s_1 + s_2 = 4 \text{ km}$ and $t_1 + t_2 = 4 \text{ min}$, so using (ii) and (iv),

$$\frac{v^2}{2} \left(\frac{1}{a_1} + \frac{1}{a_2} \right) = 4 \dots (v)$$

$$\text{and } v \left(\frac{1}{a_1} + \frac{1}{a_2} \right) = 4 \dots (vi)$$

Dividing eqn. (v) by eqn. (vi), we get $v = 2$

Putting this in eqn. (vi),

$$\frac{1}{a_1} + \frac{1}{a_2} = 2 \frac{\text{min}^2}{\text{km}} = \frac{2 \times 3600}{1000} \frac{\text{s}^2}{\text{m}}$$

$$= 7.2 \text{ m}^{-1} \text{ s}^2$$

45. (a) $a \neq 0$

Explanation: The negative value of acceleration signifies deceleration or in other words, the velocity is decreasing.

46.

(b) 1 ms^{-2}

Explanation: Given,

$$u = 18 \text{ km/h} = 18 \times 1000 / 60 \times 60 = 5 \text{ m/s}$$

$$v = 36 \text{ km/h} = 36 \times 1000 / 60 \times 60 = 10 \text{ m/s}$$

$$t = 5 \text{ sec}$$

acceleration, $a = ?$

1st equation of motion,

$$v = u + at$$

$$10 = 5 + 5a$$

$$5a = 5$$

$$a = 5/5 = 1\text{m/s}$$

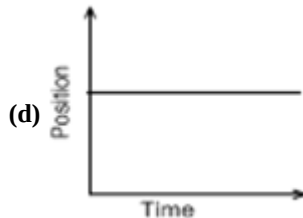
47. (a) Its position doesn't change with time with respect to the observer.

Explanation: When a body does not change its position with respect to time, it is said that body is in a state of rest. Rest and motion are relative to the observer.

48. (a) Position and acceleration as ball B

Explanation: Position and acceleration of ball A is same as that of ball B at the instant when ball A passes ball B.

49.



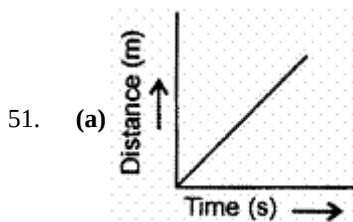
Explanation: The above graph shows an object which is not moving i.e. it is at rest.

The straight horizontal line parallel to the time axis shows that the distance covered by the body remains constant with the change in time.

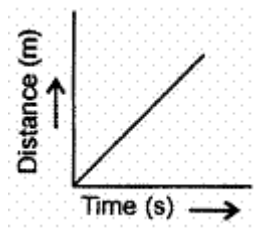
50.

- (c) zero displacement

Explanation: When the body comes back to its starting point, it has zero resultant displacements but covers a certain non-zero distance.



Explanation: Uniform motion of a moving object



52.

- (d) Vector

Explanation: Any quantity that needs to be fully described by identifying its magnitude and direction is referred to as a vector quantity. By magnitude, we mean the size of the quantity, such as length or strength. By direction, we mean where the vector is pointing or where it is being directed, such as left or right, north, south, east, or west, or even up or down.

53.

- (c) uniform acceleration

Explanation: From second equation of motion, $s = ut + \frac{1}{2} at^2$

if object starts from rest i.e., initial velocity(u) = 0 and acceleration(a) in time (t)

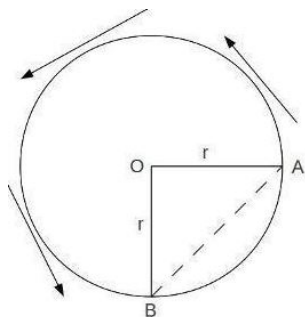
$$\text{Then, } s = 0 \times t + \frac{1}{2} at^2$$

$$s = \frac{1}{2} at^2$$

$$s \propto t^2, \text{ if } a = \text{constant}$$

So, the object with constant or uniform acceleration.

54. (c) A car is moving on a straight road
Explanation: The distance moved and magnitude of displacement are equal only in the case of motion along a straight line. Because displacement is the shortest path between initial and final path. So, for car moving on straight road, distance moved and magnitude of displacement are equal.
55. (b) $\frac{u^2}{2g}$
Explanation: If a body is thrown vertically upwards with initial velocity "u"
Maximum Height of body = $\frac{u^2}{2g}$
As $2gh = v^2 - u^2$
At maximum height final velocity of body is zero.
 $\Rightarrow v = 0$
So, $2gh = v^2 - u^2$
 $2gh = -u^2$
 $h = -\frac{u^2}{2g}$
Distance can't be negative so, $h = \frac{u^2}{2g}$.
56. (d) Statement (ii) is correct.
Explanation: Statement (ii) is correct.
57. (d) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
Explanation: (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
58. (d) 3.75 m
Explanation: Height of tap = 5 m
For the first drop, $5 = ut + \frac{1}{2}gt^2 = \frac{1}{2} \times 10t^2 = 5t^2$ or $t^2 = 1$ or $t = 1$ s. It means that the third drop leaves after one second of the first drop, or each drop leaves after every 0.5 s. Distance covered by the second drop in 0.5 s
 $= \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (0.5)^2 = 1.25$ m
Therefore distance of the second drop above the ground = $5 - 1.25 = 3.75$ m.
59. (d) The velocity of car P is increasing at a decreasing rate from 40 s to 45 s in same direction.
Explanation: From 40 s to 45 s, direction of car P is same but velocity is increasing at a decreasing rate of acceleration.
60. (d) distance travelled
Explanation: The area under a velocity-time graph represents the distance covered and the gradient of a velocity-time graph represents the acceleration.
61. (b) Both Negative acceleration and Retardation
Explanation: If there is a decrease in acceleration, it is called Retardation. This means the rate of decrease in velocity is called **retardation or negative acceleration or deceleration**.
62. (d) displacement = $\sqrt{2}r$ Distance = $\frac{3\pi r}{2}$
Explanation: Distance covered is $\frac{3}{4}$ circumference.
 $\frac{3}{4} \times 2 \times \pi \times r = \frac{3\pi r}{2}$
Displacement = Shortest Path



If We Join Point A and B we get right angle triangle with Hypotenuse(displacement) and other two sides of a triangle with sides r.
 Displacement = $\sqrt{r^2 + r^2} = \sqrt{2r^2} = \sqrt{2}r$

63.

(b) (A)

Explanation: If there is a decrease in acceleration, it is called Retardation. This means the rate of decrease in velocity is called Retardation. The negative acceleration is termed retardation or deceleration.

SI unit of acceleration (a) is m/s^2 or ms^{-2} .

So, the unit of retardation is $-\text{ms}^{-2}$.

64.

(d) A, B and D

Explanation: A body at rest has an initial velocity equal to 0.

A body that comes to rest has a final velocity equal to 0.

Acceleration is defined as rate of change of velocity. A body travelling with uniform velocity, has no change in its velocity. Hence, the acceleration is zero.

A body under free fall has an acceleration of 9.8 ms^{-2}

65. (a) (A)

Explanation: If velocity is constant then the speed has to be constant. A body can have a constant speed but a changing velocity because the direction can change while the speed is constant. (A car goes 50 mph around a roundabout). However, a body cannot have a constant velocity at changing speed.

66. (a) (A) and (D) are correct

Explanation: In a uniform circular motion, the direction of the body continuously changes, and hence the velocity (speed) of the body also changes.

67. (a) A and D

Explanation: Uniform circular motion can be described as the motion of an object in a circular path at a constant speed. When a body moves along a circular path then its direction of motion keeps changing continuously.

68.

(d) C and D are correct

Explanation:

- When a body moves from one position to another, the shortest (straight line) distance between the initial position and final position of the body is known as its displacement. It's a vector quantity.
- Displacement can be zero, less than the magnitude of distance but can never be more than the magnitude of distance covered.

69.

(d) (A)

Explanation: Speed and distance are scalar quantities because it can measure the only magnitude. But velocity and displacement are vector quantities because it can measure both magnitude and direction.

70.

(d) When a body moves with constant speed its acceleration is zero

Explanation: If a body with constant speed is travelling in the same direction (i.e. it is not changing its direction) then its

velocity is constant and so its acceleration will be zero. But if the object is changing its direction then its velocity is also changing and so it possesses the acceleration.

Hence, the given statement is incorrect.

71.

(d) (A), (B) and (D) are correct

Explanation: A body starting from rest and when it comes to rest has velocity equal to zero.

A body moving with uniform velocity has zero acceleration since there is no change in velocity.

A body under free fall has an acceleration of 9.8 ms^{-2} .

72.

(b) (A)

Explanation: Motion can be non-uniform if there is a change in velocity.

73.

(b) 2 second

Explanation: $u = 0 \text{ m/s}$, $t = 4 \text{ sec}$, $a = 9 \text{ m/s}^2$ (acceleration due to gravity)

$$s = ut + \frac{1}{2}at^2 = 0 \times 4 + \frac{1}{2} \times 9 \times 4 \times 4 = 72 \text{ m}, \frac{1}{4} \text{ of } 72 = 18 \text{ m}$$

$$\text{Now, } s = 18 \text{ m}, u = 0 \text{ m/s}, a = 9 \text{ m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$18 = 0 \times t + \frac{1}{2} \times 9 \times t^2$$

$$18 = \frac{1}{2} \times 9t^2$$

$$t^2 = 2 \times 2$$

$$\text{Hence } t = 2$$

74.

(b) 16 metre

Explanation: Here total time taken by the particle = $T = t_1 + t_2 = 4 \text{ sec}$.

Where t_1 = time during when the particle accelerates

t_2 = time during which the particle deaccelerates.

The maximum velocity attained = 8 m/sec

$$S = \frac{1}{2}vt = \frac{1}{2} \times 8 \times 4 = 16 \text{ m}$$

75.

(d) 0 kmh^{-1}

Explanation: We know that, $Velocity = \frac{\text{Displacement}}{\text{Time taken}}$

Now, Displacement = Final Position – Initial position

Since the car comes back to its starting point, so, displacement = 0

$$\text{Therefore, } Velocity = \frac{0}{3.3 \text{ hr}} = 0 \text{ km hr}^{-1}$$

So, the average velocity for the entire trip would be zero, because its final position and initial position are identical.